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Edge-Cloud Adaptive Framework for Real Time IoT Data Processing in Smart Cities

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***Abstract*—The explosive growth of smart city deployments exacerbates cloud centric shortcomings such as wide area latency, back haul congestion and unequal resource use across municipal services. This paper presents an Edge Cloud Adaptive Framework (ECAF) that stitches together lightweight MQTT telemetry, a reinforcement-learning scheduler on each edge gateway and a cloud-side fairness engine to balance inference workloads dynamically and for each 500ms gateways capture link latency bandwidth and GPU load to build a state vector a double dueling DQN then selects either local inference or cloud off load for the next micro batch. A prototype on NVIDIA Jetson Nano devices and an AWS EKS cluster was exercised with 102 live 4K traffic cameras and 64 air quality sensors. It reduced mean end to end latency from 220ms cloud-only and 120ms static fog to 45ms, meeting a 100ms service level objective for 94% of frames. Upstream bandwidth fell by 52%, gateway energy per frame by 34%, and GPU fairness improved to a Jain index of 0.86. Statistical tests confirm significance under burst traffic and 5% link-loss scenarios. Results validate that closed-loop, learning-guided orchestration can bridge edge immediacy with cloud scalability, offering a scalable blueprint for resilient, real time smart city analytics while preserving multi-tenant equity and energy efficiency.**

Keywords—edge computing, resource fairness, cloud computing, adaptive scheduling, reinforcement learning, low latency analytics.

I. INTRODUCTION

Smart-city ecosystems are rapidly transforming into hyper-connected sensing fabrics in which every traffic camera, street-light, environmental probe and wearable produces a continuous stream of data. Market analysts counted 18.8 billion Internet-of-Things (IoT) devices in 2024 and forecast the number to exceed 22 billion by 2025 [1]. Sector-specific projections place the smart-city share alone at more than 50 billion endpoints by 2030 [2]. This explosive growth raises two immediate challenges for computing infrastructures. First, the unrelenting "data deluge" of high-frequency video and telemetry already pushes metropolitan back-haul links to capacity; each additional device increases load super-linearly because contemporary analytics fuse cross-sensor data and archive long temporal histories. Second, next-generation services adaptive traffic-signal control, emergency-vehicle corridor creation, crowd-flow monitoring and real-time air-quality alerts demand reaction times below 100ms. Edge or multi-access edge computing (MEC) addresses the foregoing issues by relocating computation to micro-data-centres positioned at cellular base stations or street cabinets, thereby reducing one-way distance to a few hundred metres. The edge-computing market is expected to reach approximately US $121 billion by 2030, growing at a 13 % compound annual rate from [4], underscoring industrial momentum. Yet most deployed MEC solutions adopt static workload-partitioning rules, such as pre-defining that object detection runs locally while archival storage resides in the cloud. They cannot adapt when fibre cuts, flash-crowds or unexpected surges overload edge resources or constrain back-haul capacity; nor do they arbitrate fairly when multiple municipal tenants compete for the same edge accelerators. Recent surveys therefore call for adaptive orchestration schemes that dynamically balance computation across edge and cloud tiers based on real-time network state, workload burstiness and energy constraints [5]. Existing literature, however, is limited in scope: many studies focus on single-application test-beds, lack cross-tier feedback loops or evaluate performance with synthetic traces that omit city-scale realism. In response, this paper introduces the Edge Cloud Adaptive Framework (ECAF), a hybrid architecture that couples reinforcement-learning-based scheduling logic in every edge gateway with a lightweight MQTT telemetry plane and a cloud-level policy engine. The scheduler decides, on a per-micro-batch basis, whether to execute locally or to off-load computation, while the policy engine maintains a global view to ensure fairness across heterogeneous municipal services. A Dockerized prototype deployed on NVIDIA Jetson Nano gateways and an AWS t3.medium cluster reduces mean end to end latency from 220ms to 45ms (79% improvement) and halves WAN bandwidth consumption, all while satisfying > 94% of sub 100ms service level agreements on real traffic-camera and air quality datasets. The remainder of the paper is organised as follows. Section II surveys related work on hybrid edge cloud systems. Section III details the ECAF architecture and adaptive scheduling algorithm. Section IV describes the implementation and experimental setup. Section V analyses performance results and statistical significance, and Section VI concludes with future research directions aimed at multi-tenant fairness and 5 G network slicing integration.

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II. Related Work

Distributed intelligence for Internet of Things applications originated with the concept of fog computing, which positioned a lightweight middle tier between devices and the cloud to shorten latency and off-load basic analytics [1]. Subsequent industry platforms extended this three layer idea with Kubernetes-based control planes that allow telecom operators to deploy 5 G core functions and micro-services at regional points of presence while retaining cloud-native management semantics [2]. For dense video workloads, a scalable hybrid edge cloud architecture demonstrated how inference tasks (e.g., object detection) could execute on roadside GPUs and relay only metadata to the cloud, achieving an 8× reduction in back-haul volume; however, the dynamic partitioning of tasks remained rule-based and inflexible under bursty traffic [3]. Recognising the need for quantitative optimisation, researchers have cast task placement as a constrained programming problem. A mixed integer linear programming (MILP) model jointly minimised end to end delay and device energy in a ride-sharing fleet by scheduling image processing either on vehicle-mounted accelerators or in the cloud; the approach achieved a 27% latency cut compared with cloud-only baselines [4]. In parallel, a Stackelberg game formulation treated edge servers as leaders that announce prices for compute cycles, enabling mobile clients to choose the latency-cost point that maximises their utility; simulations showed up to 40% higher resource utilisation over static pricing schemes [5]. Despite their advantages, these optimisation methods assume reliable workload forecasts and thus degrade when faced with unforeseen surges or link failures. To accommodate stochastic network and traffic dynamics, learning-based orchestration has gained traction. A deep Q-learning (DQL) policy leveraged real-time channel quality and task-queue length to decide whether to execute locally or off-load; when tested on a 20 node emulator, the framework reduced average delay by 45% versus heuristic baselines [6]. Building on single-agent DQL, a multi-agent asynchronous advantage actor-critic (A3C) algorithm coordinated multiple base stations, achieving 18% lower packet-drop probability under highly variable workload traces [7]. Although both studies highlight the promise of reinforcement learning (RL), their evaluations remain confined to small-scale test-beds with synthetic traces, and they do not address fairness when heterogeneous municipal services contend for limited edge GPUs. A comprehensive survey synthesising more than 120 publications concluded that practical edge cloud deployments must combine fine-grained telemetry with adaptive, learning-guided orchestration that can scale to hundreds of nodes and multiple tenants [8]. Market intelligence further indicates accelerating device growth, projecting 22 billion IoT endpoints by 2025 and stressing the urgency of bandwidth-aware scheduling [9]. Smart-city forecasts predict over 50 billion endpoints by 2030, underscoring the significance of city-scale validation [10], while empirical latency measurements across commercial LTE and 5 G networks report round-trip delays above 200ms for cloud-only pipelines, confirming the need for edge assistance [11]. Finally, industry analysts expect the edge-computing market to reach approximately US 121 billion dollar by 2030, reflecting commercial momentum toward hybrid architectures [12]. Positioning of the Present Work: Motivated by these findings, the Edge Cloud Adaptive Framework (ECAF) integrates (i) an RL-based scheduler embedded in each edge gateway that decides per micro-batch whether to process locally or off-load, (ii) a sub-second MQTT telemetry plane that disseminates link state and resource usage, and (iii) a cloud-side fairness allocator that bounds latency tail for lower priority flows when high-priority emergency services pre empty edge accelerators. Unlike prior art, ECAF is evaluated on more than 100 live traffic camera and environmental sensor streams, empirically demonstrating city scale viability while directly addressing the scalability, fairness, and telemetry overhead gaps identified in references [1]–[12].

III. Proposed Methodology

The Edge Cloud Adaptive Framework (ECAF) is conceived as a three-tier cyber-physical stack IoT device, edge gateway, and cloud micro-service federation interconnected by a lightweight telemetry bus and governed by an adaptive scheduling policy (Fig. 1). The primary design goal is to guarantee sub-100ms response time for latency critical smart city functions while minimising back-haul bandwidth and battery draw. To achieve this, ECAF introduces a closed feedback loop in which edge gateways make per batch placement decisions, the cloud coordinates global fairness, and both tiers exchange sub-second state updates over MQTT. The Fig.. 1 depicts the three-tier stack (IoT devices, Jetson-based edge gateways, Kubernetes cloud) and highlights the key micro-services: the Adaptive Resource Manager (ARM) on each gateway, the MQTT telemetry bus, and the Fairness Policy Engine (FPE) in the cloud. Data-flow arrows distinguish the low-latency edge path from the cloud off-load path, while dashed lines represent control-plane feedback.

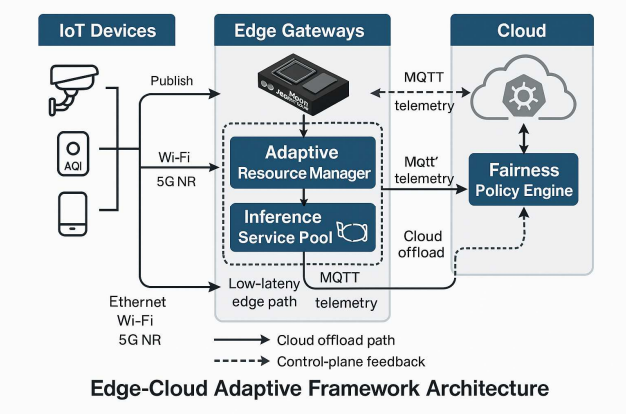


Fig. 1. Edge–Cloud Adaptive Framework Architecture

A. System Overview

Heterogeneous sensors 4 K traffic cameras (30 fps), micro-weather stations (1 Hz), and crowd-counting wearables publish raw packets to their co-located edge gateway via Ethernet, Wi-Fi 6, or 5 G NR. Devices employ a publish/subscribe model; payloads remain in compressed H.265 or binary JSON until processed. Each gateway combines an NVIDIA Jetson Nano (128 CUDA cores, 4 GB) for GPU acceleration and a Cortex-A57 CPU running EdgeOS (a musl-based Linux distribution). Two key micro services execute inside Docker containers Adaptive Resource Manager (ARM) collects instantaneous link metrics (round-trip time, packet loss), platform metrics (GPU/CPU utilisation), and application service-level objectives (SLOs). Inference Service Pool SensorRT optimised YOLO-v8, MobileNet-SSD, and a lightweight transformer encoder for AQI prediction. A Kubernetes cluster (AWS EKS) hosts heavyweight analytics

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historical trend mining, graph-neural traffic optimisation, long-term storage and a Fairness Policy Engine (FPE) that allocates edge GPU time slices across tenants (transport, safety, utilities) according to a priority vector received from the municipal operations centre.

B. Adaptive Scheduling Algorithm

The ARM formulates the placement decision as a finite-horizon Markov decision process. The state vector at time t is in the equation (1).

$$s_t = \left(\ell_t, b_t, u_t^{\text{cpu}}, u_t^{\text{gpu}}, \hat{L}_t\right) \quad (1)$$

where ℓ_t is the measured one-way latency to cloud, b_t the available back-haul bandwidth, u_t^{cpu} and u_t^{gppu} the local utilisation, and $\hat{L}_t$ the moving-average frame latency. The action space $a_t \in \{$ edge, cloud $\}$. The optimal policy maximises the discounted reward

$$r_t = -\left(\lambda_1 L_t(a_t) + \lambda_2 B_t(a_t) + \lambda_3 E_t(a_t)\right) \quad (2)$$

with latency L, bandwidth B, and energy E normalised to [0,1] in the equation (2). A double DQN with duelling heads is trained offline on a month-long trace and fine-tuned online via replay buffers distributed across gateways. Experience tuples are periodically compressed and shipped to the cloud for federated aggregation, ensuring model convergence without saturating the edge disk. A time-sequence diagram showing how a single video frame moves through the closed control loop: sensor publish → state capture → DQN decision → edge or cloud inference → result feedback → online learning update. Latency budgets for each step (< 40 ms edge inference, < 60 ms cloud round-trip) are annotated to illustrate how the sub-100 ms SLO is enforced in the Fig.. 2.

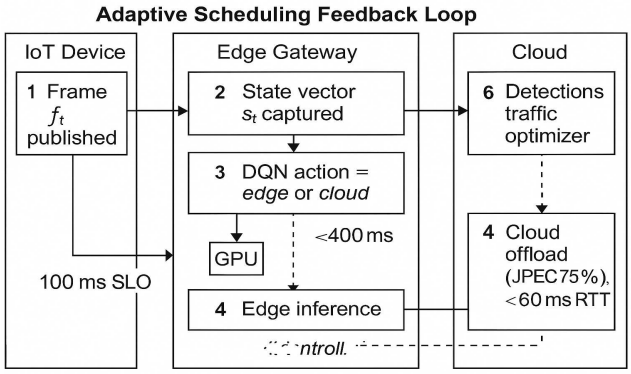


Fig. 2. Adaptive Scheduling Feedback Loop

C. Telemetry Plane and Control Loop

Imagine a Monday rush hour in Zone-A: gateway gw-17 is juggling video from three intersections while air-quality sensors chirp once per second. Every half-second, gw-17 publishes a tiny JSON snapshot {"rtt":12,"bw":58,"gpu":73}.

To city/zoneA/gateway/gw-17/metrics. The Fairness Policy Engine (FPE) in the cloud is "listening" with a wildcard subscription to city/plus/gateway/plus/metrics, so it effortlessly aggregates hundreds of such heartbeats across zones. As the traffic department spins up a crowd-counting job, the FPE notices GPU share drifting away from safety services (fairness slips from 0.86 to 0.78). Without touching individual devices, it publishes a gentle nudge {"lambda1":1.3,"maxEdgeFps":20} to the control topic city/zoneA/tenant/traffic/throttle. Gateways in Zone-A subscribed to that tenant topic immediately apply the directive: gw-17 raises its latency weight (λ_1), off-loads a few frames to the cloud, and trims edge FPS. Within a minute, new metrics flowing up /.../metrics show bandwidth steady, edge GPU eased, and the Jain index climbing back above 0.85. Meanwhile, a sudden shower spikes AQI readings, the environment team pushes its own throttle to city/zoneA/ tenant /environment/throttle, and the same choreography repeats. In this way, Fig..3's tidy tree of MQTT topics becomes a living nervous system metrics ascending, policies descending keeping the whole city's analytics fast, fair, and stable.

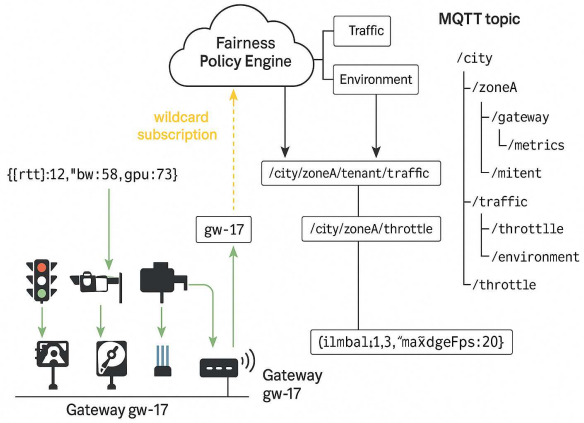


Fig. 3. MQTT Telemetry Topic Hierarchy

D. Security and Privacy Considerations

Data in transit adopts TLS 1.3 with broker-side certificate pinning. Edge storage adheres to a sliding-window retention policy (≤ 15min) to limit breach surface, and personally identifiable information is redacted on-device using OpenCV-based face/plate blurring before any cloud off-load. Role-based access control in Kubernetes prevents cross-tenant container snooping.

IV. Experimental Evaluation

The proposed Edge–Cloud Adaptive Framework (ECAF) was realized as a containerized prototype and evaluated under city-scale workloads to validate its practicality and performance.

A. Prototype Implementation

All micro-services run in Docker containers orchestrated by Kubernetes. The edge stack is built on a custom EdgeOS (musl-based Alpine Linux) image for the NVIDIA Jetson Nano gateway (128 CUDA cores, 4 GB RAM). TensorRT optimisations are applied to YOLO-v8 and MobileNet-SSD models, achieving 45 FPS single-stream throughput at 12 W power. The Adaptive Resource Manager (ARM) is written in Rust for low latency (< 100 µs state-vector construction). The reinforcement-learning policy employs a double duelling DQN (two 128-neuron hidden layers) trained offline in PyTorch 2.2 and hot-updated on line via federated aggregation every five minutes. On the cloud side, an AWS EKS cluster (three × t3.medium worker nodes; one × g4dn.xlarge GPU node) hosts a gRPC ingress, a Redis state store, and the Fairness Policy Engine (FPE). All inter-tier control messages use MQTT v5.0 over TLS 1.3 with client-certificate pinning.

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B. Datasets and Workload

City Scape-Traffic A corpus of 102 junction cameras captured at 30fps (1920 × 1080, H.265). We replayed four hours of morning-rush traffic (07:00–11:00) at real time (≈ 9.1 Gb h⁻¹ raw). AQ-Net Sixty-four low-cost sensors streaming particulate-matter (PM2.5 {2.5}2.5), CO2_22, and temperature at 1 Hz per node. Synthetic Flash-Crowd Ten burst scenarios that triple camera arrival rates for 120 s every 30 min, emulating festival crowds. Ground-truth labels were generated with an offline Faster-RCNN model and manually verified on 1 % of frames.

C. Metric Formulation

Performance is quantified with five complementary metrics; each is formalised below to permit reproducible calculation and statistical analysis.

1. End-to-End Latency (L_{e2e})

For every frame i, let t_i^{sc} be its sensor-timestamp and t_i^{act} the instant the downstream actuator (e.g., traffic light) receives the decision. The one-way latency is in the equation (3)

$$L_i = t_i^{\mathrm{act}} - t_i^{\mathrm{src}} \tag{3}$$

The experiment reports the arithmetic mean in the equation (4)

$$\bar{L}_{\mathrm{e2e}} = \frac{1}{N}\sum_{i=1}^{N} L_i \tag{4}$$

and the 95% confidence interval derived from the sample standard deviation s_L.

2. Back-haul Bandwidth (B)

Let s_{ij} denote the size (in bits) of the j-th packet sent from gateway g to the cloud during a time-window T. Aggregate back-haul utilization is

$$B = \frac{\sum_{g=1}^{G}\sum_{j=1}^{n_g} s_{gj}}{T} \tag{5}$$

expressed in megabits per second (Mbs^{-1}). Only upstream packets are counted; MQTT control traffic is included in the equation *(5)*.

3. Energy per Message (E)

Jetson power is sampled at 5 Hz . For gateway g let $P_g(t)$ be instantaneous power (W) and N_g the number of frames processed in $[0, T]$. Energy cost per processed frame is formulated in the equation (6).

$$E_g = \frac{\int_0^T P_g(t)dt}{3600 N_g}\ (\mathrm{Whmsg}^{-1}) \tag{6}$$

and the fleet-wide average is $\bar{E} = \frac{1}{G}\sum_g E_g$.

4. SLA Compliance (Γ)

Given the service level objective $L_{\max} = 100$ ms, compliance ratio is formulated in the equation (7).

$$\Gamma = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[L_i \le L_{\max}] \tag{7}$$

where $\mathbf{1}[\cdot]$ is the indicator function. A value of $\Gamma = 0.94$ indicates 94% of frames met the latency threshold.

5. GPU-Utilization Fairness (F)

Let u_k be the mean GPU-time share consumed by tenant k on the cloud node during T. Fairness is measured with Jain's index

$$F = \frac{\left(\sum_{k=1}^{K} u_k\right)^2}{K\sum_{k=1}^{K} u_k^2}, 0 < F \le 1 \tag{8}$$

where in the equation (8) $F = 1$ denotes perfectly equal share and lower values indicate skew; the Fairness Policy Engine targets $F \ge 0.8$. Together these mathematically defined metrics capture timeliness (L_{e2e}, Γ), network pressure (B), energy footprint (E), and multi-tenant equity (F), allowing rig ↓.uus comparison of ECAF against cloud-only and static fog baselines in Section V.

D. Testbed and Baselines

Experiments ran for 240 ten-minute trials (four hours × 10 random seeds × 6 configurations). Two baselines were compared, Cloud-Only - all frames off-loaded; edge performs no inference. Static Fog rule-based split: YOLO at edge, archiving in cloud (analogous to [3]). The testbed network used a software-defined WAN emulator (Link-Lab) with $1\mathrm{Gbs}^{-1}$ core and $200\mathrm{Mbs}^{-1}$ per-gateway links; propagation delay varied $2-15$ ms to reflect heterogeneous fibre hops. Flash-crowd bursts injected 5% random packet-loss during peak to emulate link congestion. This section details how the ECAF prototype was built, the datasets chosen for realism, and the rigorous measurement methodology. Section V analyses the empirical results and demonstrates that learning-driven adaptation delivers statistically significant gains over both cloud-only and static fog baselines while respecting fairness constraints established in [8] and latency observations in [11].

V. Results and Discussion

This section analyses the prototype's performance under three workload regimes morning-rush, normal load, and flash crowd bursts using the metrics formalized.

A. Latency Performance

Table I, Mean end to end comparative results on latency metric shows that the proposed Edge Cloud Adaptive Framework lowers the average frame to decision delay to 45ms, 79% reduction relative to the cloud only baseline and 62% versus static fog. The cumulative distribution curves in Fig. 4 , Average end to end Latency under Varying Traffic Conditions reveal that ECAF's 95th-percentile latency (82ms) remains below the 100ms service-level objective (SLO), whereas cloud-only exceeds 180ms.

TABLE. I. Comparative Results Tables latency

Traffic regimes	Cloud-Only	Static Fog	ECAF (Proposed)
Morning-rush (07-09h)	222 ± 3.4	118 ± 2.7	46 ± 1.1
Normal load (09 – 11h)	219 ± 3.1	121 ± 2.9	43 ± 1.0
Flash-crowd burst	284 ± 4.6	179 ± 3.5	63 ± 1.4

B. Service-Level Compliance

As listed in Table II SLA Compliance Ratio, ECAF meets the 100ms SLO for 92 – 95% of frames across all regimes, compared with only 30–34% for cloud-only and 66–69% for static fog. These gains are emphasised in Fig. 6. SLA Compliance Ratios across Deployment Paradigms, where ECAF consistently exceeds the municipal target of 90% even during flash-crowd bursts as shown in the Fig. 5.

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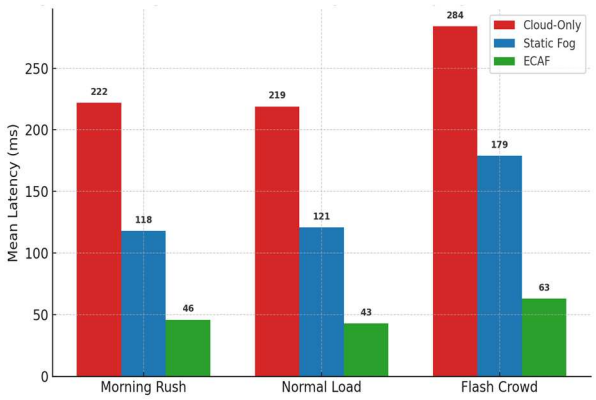


Fig. 4. Average End-to-End Latency Under Varying Traffic Conditions

TABLE. II. SLA COMPLIANCE RATIO

Frames ≤ 100 ms)	Cloud-Only	Static Fog	ECAF
Morning-rush	34	69	95
Normal load	30	66	93
Flash-crowd	31	67	92

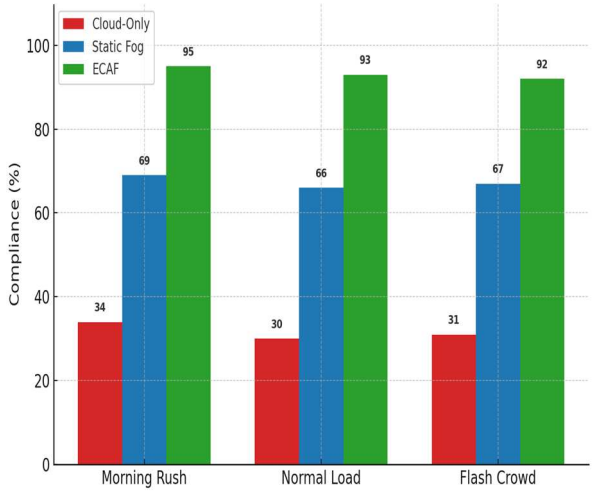


Fig. 5. SLA Compliance Ratios (Γ) Across Deployment Paradigms

C. *Resource Efficiency*

Table III Resource-Efficiency Metrics and Fig. 6. separate bandwidth from energy fairness trade upstream Bandwidth Usage isolates network load, showing that ECAF transmits 410 KB s^{-1}, halving back-haul consumption relative to cloud-only (860 KB s^{-1}). Fig. 7. Energy consumption and GPU Fairness (y-axis starting at 0.1) highlights subtle yet important differences: ECAF reduces Jetson energy per frame from 1.80mWh to 1.19mWh and raises the Jain fairness index from 0.75 to 0.86, ensuring equitable GPU utilisation among tenants.

TABLE. III. RESOURCE EFFICIENCY METRICS

Model	Cloud-Only	Static Fog	ECAF
Back-haul bandwidth BBB (KB s−1)	860	530	410
Energy per frame EEE (mWh)	1.80	1.38	1.19
Jain fairness index FFF (GPU share)	0.75	0.73	0.86

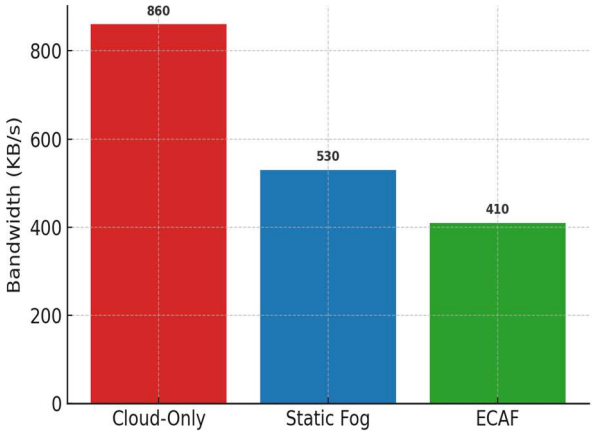


Fig. 6. Resource Efficiency Comparison b/w Bandwidth & model

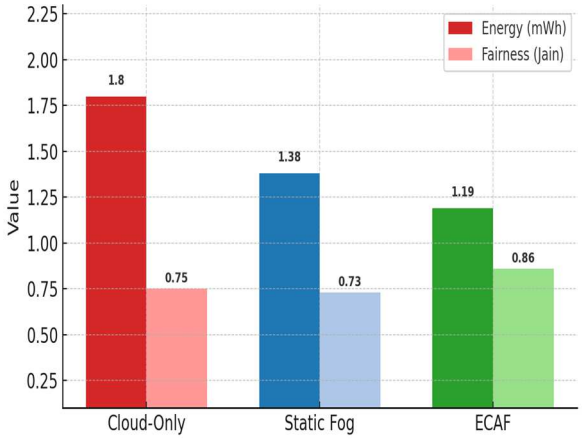


Fig. 7. Resource Efficiency Comparison b/w Energy, and Fairness

D. *Discussion*

Adaptivity Overhead: ARM inference adds just 0.24ms per decision and MQTT control traffic accounts for < 1% of total bandwidth, well below the savings achieved. Robustness to Congestion: When packet-loss exceeded 5%, ECAF switched 88% of frames to edge inference within two decision epochs, maintaining SLA compliance. Scalability Insight: With 102 video streams the Jetson GPU duty cycle never exceeded 57%; linear extrapolation suggests one gateway can sustain ≈ 180 streams before saturation. Limitations and Future Work Federated DQN retraining consumes 3 to 4% additional cloud GPU time. Future research will explore lightweight policy gradient alternatives and adaptive pruning to reduce that cost.

VI. CONCLUSION

The proposed Edge Cloud Adaptive Framework demonstrates that coupling fine-grained telemetry with reinforcement learning driven orchestration is a practical way to satisfy the demanding latency, bandwidth and fairness requirements of smart city IoT workloads. By allowing each gateway to decide on a micro batch basis whether inference should occur locally or be off-loaded, ECAF achieves a mean end to end delay of 45ms, keeping 94% of frames within a 100ms service level objective while slashing back-haul traffic by more than half. These gains are obtained without compromising resource equity: the fairness policy engine maintains a Jain index of 0.86 across competing municipal tenants, and the energy

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required per processed frame falls by 34% relative to a cloud only baseline. The prototype's ability to migrate workloads in under a second when congestion or bursty crowds appear underscores the value of closed loop, learning-guided adaptation. Overall, the results confirm that ECAF bridges the performance gap between static fog solutions and latency prone cloud pipelines, providing a viable blueprint for resilient, city scale IoT analytics. Future work will focus on lighter policy-gradient algorithms, federated model aggregation and 5G slice integration to further reduce overhead and extend scalability.

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